

**From Proto-Agent to Person:
A Bootstrapping Architecture for the Development of
Personhood**

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Abstract

A human newborn has homeostatic regulation, affect, sensorimotor coupling, social-orienting biases, and a capacity for statistical learning, but no language, no explicit norms, and no autobiographical self. By the age of three to five a recognizable individual has formed, and by adulthood the same initial architecture can support a scientist, a citizen, a parent, or a legal subject. We propose a bootstrapping architecture for this divergence. Taking as given the interactionist consensus of neuroconstructivism and developmental systems theory, the account contributes a specific staging: biologically prepared proto-agential systems enter caregiver-mediated loops of affect regulation, joint attention, language, norm enforcement, autobiographical narration, executive scaffolding, and cultural-tool use, and the child internalizes regulatory and symbolic structures that others first run around it. Each acquired layer changes the learning conditions for the next, and control migrates from a multi-person system into a single agent. The synthesis draws on attachment theory, predictive processing, language-acquisition research, Vygotskian internalization, shared intentionality, autobiographical-memory theory, executive-function development, and cultural evolution; it extends the process past childhood into literacy, schooling, law, and science; and it is checked against institutional deprivation and late first-language acquisition, where the upper layers fail and the proto-agent persists. Because an integration of this breadth accommodates most of its evidence by construction, we separate consistency claims from a smaller set of mediation and ordering predictions on which the architecture can be tested. The account is developmental and cognitive: the newborn lacks adult cognitive personhood and has full moral standing. The developmental layering parallels the historical layering of the person concept described in classical anthropology.

1. Introduction

The same newborn brain can become a non-linguistic survivor of extreme deprivation, an ordinary social adult, or a research mathematician who manipulates objects no perceptual system evolved to track. The range of outcomes is large, and the starting hardware is, to a first approximation, constant. The genome cannot account for the spread directly, because it is shared across the range, and the environment does not supply the outcome whole, because a tape recorder left running in an empty room teaches an infant almost nothing. The explanation has to be a developmental process that couples a specific kind of organism to a specific kind of surround and produces, over time, capacities present in neither alone.

Two classical answers are inadequate. Strong nativism holds that personhood unfolds from an internal program given biological time and adequate nutrition. It cannot explain why children deprived of accessible language in the first years never fully acquire grammar afterward, or why institutionally reared infants show lasting deficits in memory and executive control that nutrition does not repair. Strong environmentalism holds that culture writes personhood onto a passive surface. It cannot explain why the infant is active from the first day, recruiting caregivers, predicting contingencies, orienting to faces and voices, resisting and exploring, or why temperament, sensitive periods, and embodied constraints shape what any environment can install.

We argue that personhood is bootstrapped. A small set of biologically prepared capacities suffices to enter a structured social environment as a participant. Participation generates new learning data, the data support the acquisition of a new capacity, and the new capacity changes how the child participates and therefore which data it receives next. Language changes what can be remembered; memory changes the content of the self; the self changes which goals the child pursues; norms change how actions are selected; schooling changes the uses of attention. At each turn, the system that performs the next round of learning differs from the system that performed the last.

The rejection of both strong nativism and strong environmentalism is the standing consensus of developmental science, and a research tradition has been built on it. Karmiloff-Smith's representational-redescription account replaced fixed innate modules with a process by which the developing mind progressively re-represents its own implicit knowledge in explicit, flexible forms, dissolving the nativist-empiricist dichotomy at the level of cognitive architecture (Karmiloff-Smith, 1992). The neuroconstructivist programme of Mareschal and colleagues makes the same move at the level of brain development, treating representations as emergent products of a brain whose structure is shaped by its own activity in an environment, so that gene, neuron, body, and context act as partial constraints on one developmental trajectory (Mareschal et al., 2007). Developmental systems theory states the most radical version: Oyama argues that genes do not transmit information ready-made and the environment does not impose it ready-made, and that the whole developmental system constructs it anew in each life cycle; Griffiths and Gray generalize the point into an account of inheritance and evolution in which no single factor is privileged as the carrier of form (Oyama, 1985; Griffiths & Gray, 1994). In these literatures the interactionist conclusion, that development emerges from organism-environment coupling and from neither maturation nor instruction alone, is settled. No serious developmentalist defends the strong positions described above; they serve here as reference points, and rejecting them refutes no live opponent.

The present account makes a specific use of the interactionist thesis. The neuroconstructivist and developmental-systems literatures analyze the mechanics of organism-environment interaction in depth, but they do not stage the route from the neonate to the institutionally competent adult as a determinate sequence of named layers, and they say comparatively little about the upper end of that route: the symbolic, normative, narrative, and institutional capacities installed across adolescence and adulthood. The account makes two commitments. The first is a specific staging, the proto-agent-to-person stack of Section 2, in which each layer rewrites the learning conditions for the next and control migrates inward. The second is an integration of literatures usually kept apart (co-regulation and attachment, joint attention and shared intentionality, language as symbolic control, norm internalization, autobiographical memory, executive function, and cumulative culture) as successive phases of one bootstrapping process that continues past childhood into the institutional layers most developmental syntheses leave unaddressed. The closest mainstream accounts each treat part of this loop. Thompson's synthesis of social understanding, relationships, self, and conscience is the nearest mainstream developmental account of "the development of the person" as such (Thompson, 2006). Tomasello's theory of ontogeny derives uniquely human cognition from shared intentionality and cultural learning (Tomasello, 2019). Nelson and Fivush model au-

tobiographical memory as a social-cultural achievement assembled from language, adult memory talk, and temporal understanding (Nelson & Fivush, 2004). None of these stages co-regulation, language, norms, narrative, executive control, and cumulative culture as successive layers of one process running from the neonate to the institution-compatible adult. The contribution is therefore synthetic: an integration of known mechanisms, with no claim to a new one.

2. Conceptual distinctions

The word “person” has at least three meanings that must be kept apart. The *moral* person is the bearer of intrinsic worth and rights, which a newborn possesses in full. The *legal* person is the bearer of formal standing within an institutional order, a status that Roman law constructed as one that can be assigned, expanded, or withdrawn. The *developmental-cognitive* person is the organized capacity for self-regulation, symbolic thought, accountability, and temporally extended identity, acquired over childhood and adolescence. The account concerns the third sense only. The newborn lacks adult-style cognitive personhood and has undiminished moral standing. Keeping the senses separate blocks the most common misreading of developmental claims, which takes a statement about cognitive assembly for a statement about who deserves protection.

Within the cognitive sense, personhood is a stack of layers. The layers overlap, develop unevenly, regress under stress, and vary across cultures, but the broad direction of travel is constant: control migrates inward. The caregiver first regulates the infant from outside; the mature agent regulates itself through internalized models, language, norms, memory, and role expectations.

Layer	Minimal description	What it adds relative to the layer below
Organism	Living human body with metabolism, homeostasis, sleep, feeding, arousal, reflexive regulation	Biological viability, present before any agency
Proto-agent	Affective, embodied system that orients, seeks comfort, avoids distress, learns contingencies	Action becomes organized around needs, though not yet symbolic or self-owned
Embodied agent	Sensorimotor learner with body schema, action prediction, object tracking, locomotion	The world becomes a field of manipulable possibilities
Social agent	Participant in dyadic exchange: gaze, affect, turn-taking, attachment, joint attention	Other minds become regulators, interpreters, and targets of action

Layer	Minimal description	What it adds relative to the layer below
Language-using self	User of names, categories, pronouns, requests, negation, tense, and eventually inner speech	Experience becomes symbolically structured and publicly shareable
Narrative self	Agent who links events into a remembered, owned, temporally ordered first-person sequence	Episodic and procedural memory become autobiographical
Moral person	Agent answerable to rules, obligations, fairness, blame, praise, promises, repair	Behavior becomes conduct, interpretable through shared standards
Institution-compatible adult	Agent legible to schools, law, medicine, money, documents, professions, citizenship	Identity becomes addressable by bureaucratic, legal, and economic systems
Reflective self	Agent who models its own beliefs, motives, mortality, ideals, and place in abstract systems	The self becomes an object of explicit inquiry and revision

The minimal-self literature sharpens the two lowest transitions. Gallagher distinguishes a *minimal self*, the immediate embodied for-me-ness of experience, from a *narrative self*, the temporally extended story of who one is (Gallagher, 2000). Infants plausibly have something like the first long before they have anything like the second. The developmental question is how the narrative self is built on the minimal self, and Sections 3 to 11 trace that construction.

3. The newborn as embodied proto-agent

The newborn is a highly plastic, socially dependent, homeostatic prediction-and-control system. Over-reading its competence distorts the developmental picture as much as under-reading it.

Early development rests on a biological control architecture. The infant breathes, feeds, cycles through sleep and wakefulness, and signals distress, although caregivers supply much of this stability from outside by regulating temperature, satiation, and arousal. Affect is present as distress, comfort, interest, satiation, and startle; guilt, shame, pride, and resentment require concepts the infant does not yet hold. Sensorimotor control exists in primitive form: reflexes, sucking, orienting, early gaze, grasping. Caregiver-seeking is robust from the start, in crying, calming to touch and voice, and proximity regulation, and is best read as the infant recruiting adults as external regulators.

Two early capacities do most of the bootstrapping work. The first is a set of social-orienting biases. Newborns preferentially track face-like configurations (Johnson et al., 1991), prefer infant-directed speech (Cooper & Aslin, 1990), prefer speech to closely matched non-speech (Vouloumanos & Werker, 2007), discriminate their native language by its rhythm within days of birth (Mehler et al., 1988), and orient to biological motion (Simion, Regolin & Bulf, 2008). These biases fall short of social cognition. They constitute a sampling policy that directs the plastic learner toward the stimuli that carry social structure, namely other people. The second capacity is statistical learning. Eight-month-olds extract regularities from the speech stream, tracking transitional probabilities between syllables well enough to segment a continuous artificial language into word-like units after about two minutes of exposure (Saffran, Aslin & Newport, 1996). Phonology, word segmentation, and pattern learning are later built on this substrate.

Predictive processing offers a unifying frame for these abilities. On the free-energy formulation, an agent acts to minimize the long-run prediction error, or surprise, of its sensory states (Friston, 2010; Clark, 2013), learning contingencies through perception-and-action loops: crying brings the caregiver, sucking brings milk, a turn of the head brings a face into view. Köster and colleagues argue that statistical learning, proprioceptive and motor learning, and early physical and social understanding can be read as instances of one prediction-error-minimizing process operating over experience-expectant priors (Köster et al., 2020). The frame treats the infant as active: from the first days it samples, predicts, and acts to bring its predictions about a social world into register.

Several claims about newborns require care. Neonatal imitation, long treated as evidence of an innate matching system, is contested: a large longitudinal study failed to find reliable imitation of modeled gestures across the first nine weeks (Oostenbroek et al., 2016), a null result that has itself been disputed (Meltzoff et al., 2018), and a meta-analysis of 336 effect sizes recovered a positive pooled estimate alongside substantial unexplained heterogeneity (Davis et al., 2021). The defensible description is early sensorimotor resonance and a readiness for social learning; adult-like imitation at birth is not established. Explicit theory of mind, measured as above-chance performance on standard false-belief tasks, consolidates around the fourth year (Wellman, Cross & Watson, 2001), although infants show earlier implicit sensitivity to others' false beliefs in looking-time paradigms (Onishi & Baillargeon, 2005). Visual self-recognition, the mark of an objective self that can become an object to itself, emerges in the second year (Lewis & Brooks-Gunn, 1979; Rochat, 2003). Stable autobiographical memory, adult grammar, explicit norms, and institutional agency are absent in any mature form. The newborn is an embodied biological control system with affective priorities, social-orienting biases, plastic prediction-driven learning, and a strong structural expectation of external regulation.

4. Co-regulation and attachment

Regulation is the first thing the social environment supplies. Before the infant can regulate its own arousal, hunger, fear, or attention, the caregiver does so on its behalf, and the manner of this regulation becomes the infant's first model of how the world responds.

Affect regulation begins as a two-body process. The caregiver feeds, holds, rocks, and soothes, bringing the infant's physiological and emotional state back within tolerable bounds. Repeated across thousands of episodes, co-regulation also teaches a generalizable expectation: distress is followed by relief, signals are answered, and states that exceed the infant's own capacity can be brought down by another person. The regulation is physiological before it is psychological. Caregivers act as external regulators of the infant's autonomic and interoceptive state, and on one account the social brain grows through this allostatic co-regulation, with the caregiver anticipating and correcting the infant's internal balance until the infant can do so itself (Atzil et al., 2018). Taipale's analysis states the developmental logic: mature affective self-regulation develops by internalizing regulatory functions initially performed by the caregiver, so that the dyadic pattern of soothing becomes the individual capacity to self-soothe (Taipale, 2016). The infant first learns that regulation happens through another person and only later runs the same pattern within its own body.

Attachment theory describes what the infant learns about the regulator. Repeated patterns of response build an internal model of whether comfort is available, predictable, intrusive, or absent, and this model organizes the infant's use of the caregiver as a secure base for exploration and a safe haven in distress (Bowlby, 1969; Ainsworth et al., 1978). The model has consequences because exploration depends on it. A child who expects reliable return can leave the caregiver to investigate the world; a child who does not must spend regulatory resources monitoring an unreliable base. Caregiver sensitivity contributes without determining the outcome. A meta-analysis found a significant but modest association between parental sensitivity and attachment security ($r = .22$ across 30 studies, $N = 1,666$; the larger pool of 66 studies covered all parental antecedents combined), which leaves substantial room for temperament, context, and later relationships (De Wolff & van IJzendoorn, 1997).

The active-inference reading of attachment treats the internal working model as a learned prior over social availability. The infant forms expectations about whether signaling will reduce uncertainty and distress, and the secure, avoidant, and ambivalent patterns correspond to different policies for minimizing free energy over interoceptive stress states when the caregiver's responses are reliable, reliably unresponsive, or inconsistent (Cittern et al., 2018). This reading ties attachment to the prediction-and-action machinery that drives early learning in general and places co-regulation as the first layer of the same architecture.

The external regulator is not always benign. The loop that compiles trust, recovery, and exploratory confidence can equally compile vigilance, dysregulation, and the expectation that signals go unanswered. The architecture is neutral about the content installed. It guarantees only that some content will be installed, because the infant learns from whatever regulator it has.

5. Social address and caregiver functions

Caregivers treat the infant as a person before it is one in the cognitive sense, and this early treatment is itself a developmental mechanism. They name the infant, address it in the second person, attribute intentions and feelings to its movements, and interpret its cries as meaningful commu-

nications about a world. The infant cannot yet support these attributions, but they structure the interactions from which it learns.

Adults read intention into behavior that does not yet carry it. A reach is treated as wanting, a gaze as looking-at, a cry as asking-for. The caregiver responds to the imputed intention, and the response makes the imputed structure real in the next round: an infant who finds that reaching reliably produces the reached-for object via an adult learns to reach as a communicative act. The “someone” the caregiver addresses is at first a useful fiction, and acting on the fiction generates the contingencies that eventually make it true.

Affect mirroring is the most studied case. Gergely and Watson’s social-biofeedback theory holds that caregivers reflect infant affect in marked, exaggerated, and well-timed displays, and that the infant uses these displays as external feedback about its own internal states (Gergely & Watson, 1996). The marking is essential. The caregiver’s display is recognizably a representation of the infant’s state, distinct from the caregiver’s own feeling, and this lets the infant bind a diffuse bodily-affective pattern to a categorized, shareable emotion. On the mentalization account, marked mirroring is also the route by which a caregiver’s reading of the infant’s mind scaffolds the infant’s own affect regulation and, later, its capacity to represent mental states as such (Fonagy, Gergely, Jurist & Target, 2002). Emotional self-awareness thus begins as a function supplied from outside and is only later generated within.

Taken together, the functions caregivers perform across these early loops constitute a temporary external runtime for personhood. The adult supplies regulatory and representational operations the child cannot yet perform, and runs them in the child’s presence and on the child’s behalf until the child can emulate them. The table collects the operations that this and the following sections assign to the caregiver.

Adult function	Operation supplied from outside	What the child later internalizes
External regulator	Stabilizing arousal, hunger, fear, attention	Self-regulation
Attention director	Pointing, showing, naming the relevant	Endogenous control of attention
Symbolic interpreter	Mapping world and body into signs	Symbolic self-representation
Norm enforcer	Correction, permission, the error signal	Self-imposed accountability
Memory co-author	Storing and reconstructing early events	Autobiographical narration
Identity assigner	Name, kinship, category, family narrative	A self-description to revise
Cultural package manager	Installing tools, scripts, rituals, texts	Internalized cultural competence

Adult function	Operation supplied from outside	What the child later internalizes
Executive prosthesis	Planning, sequencing, inhibiting, reminding	Endogenous executive control

The runtime description is a heuristic with a clear limit. The brain is an embodied, self-organizing system with evolved priors, temperamental variation, and sensitive periods, and the child runs its own processes throughout: predicting, recruiting, imitating, resisting, exploring, and through its responses reshaping the environment that scaffolds it. The runtime is also indifferent to content. The loop that installs trust, curiosity, and self-command can install fear, shame, and learned helplessness, because the child internalizes the regulation it receives whether or not that regulation meets its needs. The heuristic accurately describes a transfer: operations that begin distributed across a multi-person system migrate, over years, into a single agent. Sections 6 to 11 trace that transfer one capacity at a time.

Identity assignment runs alongside the regulatory functions. The infant is given a name, a kinship position, a set of social categories, and a place in a family narrative, and is held to expectations attached to these long before it can author any of them. The child learns who it is to others as the first available answer to who it is. At the developmental base this recovers the oldest layer of personhood in the anthropological record, the self as a node in kinship, ritual, and recognition, which precedes the notion of a bounded interior given at birth (Mauss, 1985). The child is recognized into a position in a web of relations, and that position is the scaffold for a more individuated self.

6. Joint attention and shared intentionality

Long before the first words, infants engage in proto-conversational exchanges of gaze, expression, and vocal turn-taking, the primary intersubjectivity that precedes and prepares joint attention (Trevarthen & Aitken, 2001). Around the end of the first year this dyadic exchange of gaze and affect between infant and caregiver becomes triadic. The infant follows a caregiver's gaze or point to an object, checks back to the caregiver's face, and coordinates attention on a shared target. It also begins to point declaratively, directing an adult's attention to an object for the sake of shared interest, with no request for the object (Liszkowski et al., 2004). Self, other, and world now form a triangle.

Triadic structure is the platform for word learning, teaching, and shared reference. To learn that a sound names a thing, the infant must know which thing the speaker means, and joint attention solves this reference problem by aligning what speaker and learner attend to. Carpenter, Nagell, and Tomasello documented the emergence of joint-attention skills between 9 and 15 months of age and their close relation to early language, although joint attention is a platform for language and not by itself a sufficient cause (Carpenter, Nagell & Tomasello, 1998). The input must be social. Infants who received live, interactive exposure to a foreign language showed phonetic learning that audio and video recordings of matched content failed to produce, which indicates that contingent

social interaction drives early language learning beyond what acoustic exposure alone provides (Kuhl, Tsao & Liu, 2003).

Tomasello's account places shared intentionality at the center of distinctively human cognition. Beyond attending to objects alongside others, the child enters "we" structures: shared goals, joint plans, divided roles, and a sense of how a task is done that is held in common and treated as binding. The progression runs from dyadic engagement through joint attention to collaborative activity in which each participant holds a role and tracks the other's (Tomasello et al., 2005; Tomasello, 2019). Ostensive communication trains the child to treat some signals as conveying generalizable knowledge, distinct from immediate commands. Csibra and Gergely's natural-pedagogy account argues that ostensive cues (direct gaze, infant-directed speech, and contingent responsivity) mark information as communicative and generic, which allows cultural knowledge to be taught as well as demonstrated (Csibra & Gergely, 2009). These perspective-taking abilities underlie the imitative, instructed, and collaborative forms of social learning that distinguish human cultural transmission (Tomasello, Kruger & Ratner, 1993).

In the "we" layer, individual agency becomes participation in shared structure. Cooperation, teaching, role division, and norm-governed joint action all presuppose that two agents can hold a goal in common and treat its standards as obligations on both. The institutions installed by later layers build on this capacity, since an institution is a shared intentional structure stabilized across many agents and inherited across generations.

7. Language and symbolic control

Language is a control layer that reorganizes perception, memory, social life, and self-regulation. By entering a linguistic community the child acquires new forms of thought as well as labels for thoughts it already had, so that language functions as cognitive architecture in addition to serving as a communication code.

The reorganizing functions can be itemized. Naming creates stable handles for entities, actions, states, and relations, so that the world becomes addressable. Categorization compresses perception into reusable concepts, so that "animal", "tool", "danger", and "promise" support inference beyond the instance. Self-reference, through the name and then the shifting pronouns, lets the organism become an object to itself: the child learns that "I" and "you" trade places depending on who speaks, a non-trivial discovery that the self is both a subject of experience and an item others refer to. Temporal vocabulary links past, present, and future, so that "I was", "I will", and "I promised" become storable. Causal and conditional morphology ("because", "if", "should have") enables explanation and counterfactual reasoning. Mental-state terms ("want", "think", "know", "remember") give the child a vocabulary for minds, its own and others'. Inner speech converts external dialogue into a medium of private thought.

Nelson's work on the mediated mind treats this transformation as the central event of early cognitive development. Language restructures memory, concept formation, narrative, and the understanding of other minds, because it supplies a semiotic medium in which experience can be repre-

sented, stored, recombined, and shared (Nelson, 1996). Pre-linguistic cognition is substantial. The claim concerns a large class of distinctively human operations, including autobiographical recall, explicit norm reasoning, decontextualized abstraction, and public accountability, which become available only once the child can operate on symbols.

The symbolic layer extends well past reference. Once thought can be carried in language it detaches from the perceptible situation: a child who has “yesterday”, “if”, and “because” can reason about what is absent, conditional, or counterfactual, and a community that has writing can externalize that reasoning into marks that outlast the reasoner and can be inspected, copied, and corrected by others. Inner speech connects these capacities to self-regulation. The verbal self-regulation that begins as audible private speech becomes a silent medium in which the agent poses questions to itself, holds a premise while testing a consequence, and rehearses an argument before committing to it. Scientific and mathematical reasoning lie at the far end of this trajectory. Explicit definitions, public hypotheses, and the obligation to be refutable are operations on symbols beyond the reach of the bare perceptual system, available only to an agent that has internalized a language and the collective practices built on it.

Late first-language acquisition is the boundary case that fixes the claim. Mayberry and Lock studied deaf and hearing adults who differed in whether they had accessible language in early childhood, and found that those with early first-language exposure, signed or spoken, later achieved high competence in a second language, while those who lacked accessible language in the first years performed poorly even after extended later exposure (Mayberry & Lock, 2003). The critical variable is accessibility, independent of modality: sign language acquired early supports later language as well as speech does, and speech the child cannot hear supports nothing. Early accessible language is therefore developmental infrastructure, since its absence during the sensitive window cannot be fully repaired afterward. Severe language deprivation is damaging for this reason. Language is one of the central media through which human cognitive control becomes recursive, and when the window is missed the later layers lack the layer on which they build.

8. Norm internalization and moral agency

A person in the cognitive sense is an agent whose actions are answerable. Preference and aversion alone do not amount to conduct, which is behavior interpreted through shared standards. The capacity to treat one’s own behavior as answerable to such standards is acquired, like the other capacities, from outside in.

Acquisition begins with correction. Caregivers and peers prohibit, permit, praise, and blame, and the child first encounters norms as the external consequences of acts. The decisive step is the move from “this brings punishment” to “this is forbidden”, because the second generalizes beyond the immediate enforcer to a rule that holds in a context. Children make this move early and robustly. By the third year they follow norms and also enforce them on third parties, spontaneously protesting and correcting a puppet’s violations of a game’s rules when they are not themselves affected, and treating the norm as binding on anyone in the relevant situation (Rakoczy, Warneken & Tomasello, 2008). Enforcing a rule to which one is not party is strong evidence that the rule has been

internalized as a general standard, since a personal contingency would not motivate protest on another's behalf. The same norm sensitivity appears in copying. Children faithfully reproduce even causally unnecessary steps of a demonstrated procedure, an overimitation that treats the modeled sequence as the correct way to perform the task, and this high-fidelity copying separates human children from other apes (Lyons, Young & Keil, 2007; Horner & Whiten, 2005).

Thompson's developmental account links this to the broader growth of conscience, placing norm sensitivity in the same matrix as social understanding, relationships, and the self (Thompson, 2006). The self-conscious emotions of accountability (guilt, shame, and pride) develop alongside the rules and presuppose both a represented self and a standard against which to measure it; their onset is tied to the objective self that emerges with mirror self-recognition in the second year (Lewis et al., 1989). These emotions convert a violated standard into an experienced obligation. A child who feels the pull to apologize after a rupture has internalized the rule together with the stance of being answerable for breaking it.

The scope of norms continues to widen well past early childhood. Family norms give way to peer norms, peer norms to institutional norms, and institutional norms to the explicit, codified, jurisdiction-specific norms of law and profession. Each widening reuses the same machinery: a standard is encountered as external enforcement, abstracted into a general rule, bound to an emotion of accountability, and folded into the self-model as a constraint the agent imposes on itself. Moral personhood is the cumulative result, an agent whose action selection is shaped by standards it treats as binding whether or not anyone is watching.

9. Autobiographical memory and narrative selfhood

Adults co-author the narrative self before the child authors it. A young child has event memory and procedural memory but no organized, owned, temporally ordered story of its own life. That story is assembled in conversation with adults who remember on the child's behalf and who, by remembering aloud, teach the child the form a life story takes.

The mechanism is elaborative reminiscing. Caregivers revisit shared events with the child in a particular grammar: who was there, what happened, what the child felt, why it mattered, and what kind of person the child was in the event. Nelson and Fivush's social-cultural theory holds that autobiographical memory emerges across the preschool years from the convergence of basic memory capacity, language, adult memory talk, temporal understanding, and a developing sense of self and other (Nelson & Fivush, 2004). Children of caregivers who reminisce in an elaborative style, asking open questions and richly reconstructing past episodes, develop more detailed and coherent autobiographical narratives, and the longitudinal evidence suggests that the association runs at least partly from the conversational style to the child's later memory (Reese, Haden & Fivush, 1993). The child learns to remember in the first person by being guided through first-person remembering until it can proceed unaided.

Childhood amnesia is consistent with this construction. Adults retain almost no episodic memories from before roughly the third or fourth year, and the offset of the amnesia tracks the devel-

opment of language and narrative skill more closely than the development of memory storage. The co-authoring account predicts this pattern: events are encoded before the child has the narrative apparatus to organize them into a retrievable autobiography, and the later loss concerns the indexing system that would have made them tellable.

Narrative selfhood results from this process running for years. The self becomes a temporally extended protagonist, an agent who did things, to whom things happened, and who has a trajectory and a character. The family supplies the first version of the trajectory and the first reading of the character, and the child inherits both before revising them. The narratable individual, the named self with a tellable life course, depends throughout on recognition and narration by others.

10. Executive function, private speech, and inner speech

Adult agency requires capacities that the young child lacks and the caregiver initially supplies: holding a goal in mind, inhibiting a prepotent response, sequencing steps, correcting errors, and switching between tasks. These executive functions develop steeply across the preschool and school years, and their development is partly the internalization of an external supervisor.

The route is the one Vygotsky described: higher mental functions appear first between people and only later within the child, as interpsychological regulation becomes intrapsychological (Vygotsky, 1978). The adult regulates the child's action from outside through speech, the graded support that lets a child accomplish what it cannot yet manage alone and that Wood, Bruner, and Ross (1976) termed scaffolding: "wait", "first this, then that", "don't touch", "try again", "use your words". The child then reproduces this speech aloud, talking itself through tasks in audible self-directed private speech, and finally compresses the private speech into silent inner speech that guides action from within, the trajectory from egocentric to inner speech that Vygotsky (1987) made central to verbal thought. The trajectory is well attested: private speech increases when tasks are difficult, predicts better performance, and is gradually internalized into covert verbal self-regulation, so that social command becomes self-command (Winsler, Fernyhough & Montero, 2009). The supervisor that once stood outside the child comes to operate inside it, in the same verbal medium it used when it was external.

The executive layer is the clearest case of the architecture's central pattern, and it shows that the pattern describes a mechanism. A regulatory function transfers measurably from a two-person system to a one-person system, mediated by a public representational medium that becomes private. Working memory, inhibition, and cognitive flexibility have a maturational component grounded in prefrontal development, but maturation supplies only the substrate. In humans the content of self-regulation (the specific instructions the child gives itself, the order in which it does things, the standards against which it checks its work) is drawn from the social regulation it received and then run on maturing neural hardware. School formalizes and intensifies this transfer, imposing schedules, routines, and explicit demands for sustained attention and delayed action that train executive control well beyond what unstructured life requires.

11. Cumulative culture and institutional personhood

Most developmental accounts stop in childhood. The bootstrapping architecture continues, because the higher layers of personhood are installed across adolescence and adulthood and depend on resources that no individual could generate alone. The adult is a nervous system coupled to cumulative culture, and this coupling produces the capacities most characteristic of human personhood.

Cumulative culture supplies the inheritance on which these layers draw. Human populations preserve and modify innovations across generations through high-fidelity social transmission, so that each learner starts from an accumulated stock of tools, symbols, procedures, and institutions that took many lifetimes to build. This ratchet, in which faithful copying prevents loss and allows variation to accumulate, runs at a scale no other species matches (Tomasello, 1999; Tennie, Call & Tomasello, 2009), and it underlies the dual inheritance, genetic and cultural, that shaped human cognition (Boyd & Richerson, 1985; Henrich, 2016). The child does not reinvent number, writing, law, or scientific method. It inherits them in compressed, transmissible form and internalizes them through the same loop of participation followed by internalization that governed the earlier layers, now applied to cultural artifacts.

The institutional layers can be itemized by what each installs. Literacy externalizes memory into durable marks, so that a person can think with more than working memory holds and can stand in relation to texts produced by the dead. Numeracy supplies a compression system for quantity and relation that perception does not provide. Schooling shapes attention, imposing sustained focus, sequenced curricula, and the habit of treating decontextualized symbols as objects of attention. Law supplies a public grammar of personhood, defining who may own, contract, testify, and be held answerable, in direct institutional descent from the Roman *persona*. Money and identity documents make the person legible to economic and bureaucratic systems, so that the self acquires a representation that institutions can read and act on. Science installs the norms of a collective self-correcting practice: explicit hypotheses, public evidence, and the obligation to be refutable. Professions package role-specific competence, obligation, and identity into inhabitable forms. Each of these is a cultural structure first run around the developing agent and then partly internalized as a capacity and a self-description.

The same initial architecture can therefore diverge into a deprived survivor or a research scientist. The difference lies principally in the quantity and quality of external cognitive structure the agent has had the opportunity to internalize, and far less in brain size or general intelligence. Education has a measurable cognitive effect of the kind the architecture predicts: a meta-analysis of 142 effect sizes from 42 datasets and over 600,000 participants, drawing on three quasi-experimental designs, estimated that each additional year of schooling raises measured intelligence by roughly one to five IQ points, an effect that showed little sign of fading with age (Ritchie & Tucker-Drob, 2018). The adult scientist is the infant together with an internalized body of cultural structure, and the institutions are constitutive of the highest layers of the mind they help to build.

The history of the concept of the person supports reading these layers as a genuine stratification. Mauss's classic argument treats "the person" as a category assembled over the long history of societies, moving from the relational and ritual personage through the Roman juridical *persona* to the interior moral self and finally the rational, rights-bearing individual (Mauss, 1985). The developmental architecture recapitulates that stratification in compressed form. The relational layer is installed in infancy through co-regulation and social address; the narratable individual is assembled in the autobiographical self co-authored across early childhood; the juridical and rational-autonomous layers arrive with the institutional and reflective capacities installed by schooling, law, and disciplined inquiry across adolescence and adulthood. History assembles the layers across many generations, and development runs the same layering within a single life. Disassembly at either scale, historical or biographical, therefore follows the same logic: when the institutional supports are withdrawn, the layer they sustain thins while the formal category persists.

12. Evidence from deprivation

Deprivation is the closest developmental science comes to an experiment on the architecture, and it is an ethically and methodologically compromised one. The cases never isolate a single missing input, documentation is often poor, and the children involved are frequently subject to multiple overlapping harms. The evidence nonetheless converges on a single structural conclusion.

Three lines of evidence bear on the architecture. Institutional deprivation supplies the first. The Bucharest Early Intervention Project, a randomized controlled trial of foster-care placement for institutionally reared children, found that early institutional care, characterized by low responsive caregiving and impoverished stimulation, depressed cognitive development, and that placement into responsive foster care, particularly before roughly two years of age, substantially improved cognitive outcomes (Nelson et al., 2007). Follow-ups in the same cohort traced the deficits into memory and executive function (Bos et al., 2009) and into brain electrical activity, which partially normalized with earlier placement (Vanderwert et al., 2010). The independent English and Romanian Adoptees study found that deprivation lasting beyond the first six months left neurodevelopmental and mental-health effects detectable into young adulthood (Sonuga-Barke et al., 2017). The dose-and-timing structure of these results is the signature of an experience-expectant system (Greenough, Black & Wallace, 1987): the deficits track the absence of expected caregiving input and are partly remediable if the input arrives within a window. Late first-language acquisition supplies the second line, discussed in Section 7: accessible language missing in the early years cannot be fully compensated later, which marks language as infrastructure for later development (Mayberry & Lock, 2003). Language deprivation in deaf children who lack early access to sign supplies a clinical version of the same finding, with patterns of dysfluent language, reduced general knowledge, and disrupted reasoning described under the heading of language-deprivation effects (Hall, 2017).

Extreme isolation cases, the so-called feral and confined children, are the third and weakest line. They are rare, almost always confounded by abuse, malnutrition, and possible pre-existing disability, and frequently sensationalized in ways that make clean inference impossible. The best-

documented modern case is Genie, a girl discovered at thirteen after years of confinement and isolation, who acquired substantial vocabulary but never normal grammar; this outcome is consistent with a sensitive-period limit on first-language acquisition, although her history of abuse and possible prior disability confound any single interpretation (Curtiss, 1977). Reviews of the broader material warn that such cases are too poorly documented to settle questions about human nature on their own (Candland, 1993). They are best treated as weak corroboration of conclusions established by better-controlled evidence.

The three lines support a limited conclusion: the human organism expects a social, linguistic, and cultural niche for normal personhood development, and the architecture fails in characteristic ways when that niche is withheld. The evidence does not show what a human would be in the complete absence of culture. Under deprivation the proto-agent and its local adaptations remain: affect, sensorimotor skill, nonverbal memory, attachment-seeking in distorted forms, and practical problem-solving. The upper stack fails: language-mediated cognition, narrative selfhood, explicit norm reasoning, and institutional competence. This pattern of sparing and loss is the strongest support for the architecture, because a layered, experience-expectant bootstrapping system predicts it and neither pure nativism nor pure environmentalism predicts it cleanly.

13. The integrated architecture

The mechanisms above compose into one architecture. The table maps each mechanism to the external scaffold that supplies it, the capacity the child internalizes, and its contribution to personhood.

Layer	External scaffold	Internalized capacity	Contribution to personhood
Affective regulation	Feeding, holding, soothing, repair	Self-soothing, affect tolerance	Stability of agency
Attachment	Secure base, safe haven	Trust, exploration, relational priors	A social world that can be navigated
Attention regulation	Pointing, showing, gaze-following	Joint attention, relevance tracking	A world held in common
Emotion concepts	Marked affect mirroring, labeling	Emotion differentiation	A mental-state reading of the self
Language	Naming, requests, stories, questions	Categories, self-reference, inner speech	A symbolically structured self
Shared intentionality	Joint action, divided roles	“We” structures, cooperation	Participation in shared agency
Norms	Correction, praise, blame, permission	Rule-following, accountability	Moral personhood

Layer	External scaffold	Internalized capacity	Contribution to personhood
Autobiographical memory	Elaborative reminiscing	Narrative continuity	A self extended across time
Executive function	Adult planning, reminders, private speech	Inhibition, planning, self-command	Deliberate agency
Cumulative culture	Literacy, numeracy, law, science, roles	Cultural tools and institutional competence	Institution-compatible adulthood

The architecture is a loop, and its inputs, scaffolds, internalization processes, and outputs can be stated as a conceptual schema. The pseudocode below describes the developmental process at a high level and makes the recursion and the attractor structure explicit. It is a schematic description with no claim to be an executable model or a neural implementation.

Initialize newborn:

```

body          = embodied_homeostatic_system()
affect        = active(distress, comfort, interest, arousal)
world_model    = shallow, rapidly plastic
caregiver      = uncalibrated prior over availability
social_bias    = oriented to faces, voices, biological motion
language       = no lexicon, strong statistical-learning capacity
self_model     = minimal embodied self, no narrative identity
norms          = minimal
autobiography  = minimal
executive      = externally scaffolded
culture        = none internalized

```

For each developmental interaction:

```

perceive(body, world, caregiver, symbols)
predict(causes, caregiver_response, action_outcomes)
act(cry, orient, reach, gaze, vocalize, imitate, ask, protest)

if distress > self_regulation_capacity:
    signal(caregiver)
    receive(comfort, feeding, voice, repair)
    update(caregiver_prior, affect_regulation)

if caregiver marks/labels state:

```

```

    bind(bodily_affect_pattern -> emotion_concept)

if joint_attention(self, caregiver, object):
    align(attention); create triadic reference

if language_input under social contingency:
    extract(statistical_structure)
    map(words -> objects, actions, states, relations)
    learn(names, pronouns, tense, causality, mental-state terms)

if norm_feedback:
    abstract(enforcement -> general rule in context)
    bind(rule -> accountability emotion)

if autobiographical_talk:
    encode(event + emotion + agent + cause as self-in-time)

if private_speech_practice:
    internalize(external_instruction -> self_instruction -> inner_speech)

if education_or_institution:
    internalize(cultural_tools, roles, abstract_symbols)

self_model = integrate(body, name, memory, language,
                      norms, roles, projects)

# recursion: this round's new capacity changes next round's data
caregiver_and_peers.adjust_to(self_model)

```

Two attractors follow from the loop. Where the environment supplies responsive caregivers, accessible language, stable attachment figures, joint attention, play, norms with repair, education, and cultural tools, the system converges on increasing self-regulation, language-mediated cognition, narrative selfhood, norm internalization, institutional compatibility, and reflective agency. Where the environment lacks responsive care, accessible language, stable social feedback, emotional labeling, norm scaffolding, and education, the system preserves basic organism and proto-agent functions and local survival adaptations, while the upper layers (language-mediated cognition, narrative selfhood, executive and institutional capacity) fail to develop normally. The two trajectories are one process run in two niches, so the boundary cases of Section 12 count as evidence for the architecture.

14. Consistency claims, testable predictions, and limits

An integrative account must separate two kinds of claim that are easy to conflate. Most of what an assembled synthesis offers is consistency: it accommodates the findings from which it was built, and that accommodation is retrodiction. A smaller class of claims is predictive, in that the architecture commits to a relationship that its component literatures do not each separately assert and that could turn out false. The value of the account as an integration and its much weaker standing as a falsifiable theory are separate questions and are treated separately below.

The following are consistency claims. Each restates, in the architecture's vocabulary, a relationship that the studies cited earlier already report, and listing them confirms only that the synthesis is faithful to its sources. That elaborative parental reminiscing accompanies more coherent autobiographical narrative is an established finding of the reminiscing literature (Reese, Haden & Fivush, 1993; Nelson & Fivush, 2004). That the transition from private to inner speech tracks executive-function gains is the established reading of the private-speech literature (Winsler, Fernyhough & Montero, 2009). That severe early language deprivation impairs the symbolic upper stack while sparing more nonverbal capacities is the pattern already drawn from the deprivation and late-acquisition evidence (Mayberry & Lock, 2003; Hall, 2017). That more schooling accompanies measurably more abstract cognition is the finding of the education meta-analysis cited above (Ritchie & Tucker-Drob, 2018). These relationships shaped how the architecture was assembled, so presenting them as successful predictions would be circular.

A narrower set of claims is closer to predictive, because the relationships follow from the integration without being the direct content of any single assembled study. The architecture entails specific mediation structure, and mediation is harder to retrodict than correlation. If self-repair is internalized dyadic repair, the link between caregiver repair after rupture and the child's later capacity for apology and error correction should be mediated by the child's acquisition of repair scripts in private and inner speech, and should not reduce to general attachment security. If triadic reference is specifically the platform for symbolic self-development, early joint-attention quality should predict later pronoun mastery and narrative ability beyond its known correlation with vocabulary size. The layering claim, that each layer rewrites the learning conditions for the next, predicts ordering and gating effects in addition to co-occurrence: emotion-labeling density should affect later emotion regulation through the child's emotion-concept vocabulary, with no independent path, and a layer should be hard to install when the layer below it is absent. These are the claims on which the architecture can be tested. They are directional relationships expected under reasonable controls, and several remain untested in the specific mediated form the architecture requires.

The account also excludes several claims. It does not hold that infants lack moral worth; the developmental and moral senses of personhood are separated by construction. Culture does not write on a blank slate: the newborn is an active, biased, prediction-driven learner, and temperament, sensitive periods, genetics, and embodiment constrain every trajectory. Language is taken to be necessary for the higher symbolic, narrative, and accountable layers, with no claim that it is necessary for consciousness or experience. Caregivers do not install a self into passive hardware: the

child actively recruits, predicts, imitates, resists, and explores, and its signals shape the environment that scaffolds it. Feral-child cases are not probative and only weakly corroborate conclusions established elsewhere. The “external runtime” and “symbolic operating system” descriptions are explanatory heuristics for a transfer of regulatory and representational functions from a multi-person system to a single agent, and the transfer is the substantive claim.

A final limit concerns the status of the architecture. It is an integrative synthesis: it shows that mechanisms studied in separate literatures can be staged as a single recursive bootstrapping stack from the neonate to the institution-compatible adult, and it accommodates nearly all of its evidence because it was assembled from that evidence. This carries a real falsifiability cost. A framework flexible enough to absorb co-regulation, language, norms, narrative, executive control, and cumulative culture is, by that flexibility, hard to refute as a whole, and much of what reads as confirmation is consistency with studies already in hand. The architecture differs from the interactionist consensus in its specific staging and its reach into the institutional layers, and it proposes no novel mechanism; a critic may reasonably ask whether the staging carves development at real joints or imposes an order on a continuous process. The defense is modest. The synthesis is useful if its layered, experience-expectant structure organizes the evidence more economically than the alternatives and if the mediation and ordering claims above survive tests in the form the architecture requires. If those claims fail, the architecture should be revised or abandoned; the convenience of the integration is not evidence for it.

15. Conclusion

The human infant is born as an embodied proto-agent: a homeostatic, affective, sensorimotor, socially oriented, plastic prediction-and-control system with a structural expectation of external regulation. Adult cognitive personhood is bootstrapped from this starting point. The account shares with neuroconstructivism and developmental systems theory the premise that development emerges from organism-environment coupling, and its contribution is a specific staging of that coupling and its extension into the institutional layers. The infant enters caregiver-mediated loops of co-regulation, social address, joint attention, language, norm enforcement, autobiographical narration, and executive scaffolding, and across years it internalizes the regulatory and symbolic structures that others first run around it, until control that began as a property of a two-person system becomes a property of one agent. Language makes the architecture recursive, memory binds the self across time, norms make action answerable, and cumulative culture installs the institutional and reflective layers that no individual could generate alone.

In the developmental-cognitive sense, human personhood is co-regulated, language-mediated, norm-governed, narratively extended, and culturally completed agency. The newborn is a culture-completable organism, and the completion is a developmental process with identifiable mechanisms, characteristic failure modes under deprivation, and testable consequences. The layered structure of the developmental account parallels the layering recorded in the history of the person concept, and it shares that history’s vulnerability: when the supports that sustain a layer are withdrawn, the layer thins regardless of what the formal category continues to assert.

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